Chapter 7: Is Your Effect Real?

Testing Hypotheses, Measuring Certainty, Quantifying Effects

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1 Overview

> Before You Begin
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> This chapter brings together everything you need to analyse and interpret your results: hypothesis testing to determine IF there's an effect, confidence intervals to show HOW CERTAIN you are, and effect sizes to show HOW BIG it is. This is one of the most important (and most misunderstood) topics in research methods.

This chapter covers:

- Null hypothesis significance testing (NHST)
- What p-values mean (and don't mean)
- Confidence intervals and what they tell us
- Effect sizes (Cohen's d) and practical importance
- Independent samples t-test (between-subjects)
- Paired samples t-test (within-subjects)
- Complete APA reporting with all components

All concepts are illustrated using DataLab examples from your class.

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2 Part 1: The Logic of Hypothesis Testing

2.1 The Fundamental Question

In research, we often want to know whether an effect is "real" — whether our observed difference reflects something true about the world, or whether it could have occurred by chance.

Consider your DataLab memory data:

- Lab A (abstract words): $M = 5.4$
- Lab B (concrete words): $M = 6.8$

There's clearly a difference of 1.4 words. But is this difference meaningful, or could it have happened by chance? Maybe if you tested different students, the results would be reversed.

2.2 The Three Questions

When analysing results, you need to answer three questions:

1. **Is the effect real?** — Hypothesis testing and p-values
2. **How certain are we?** — Confidence intervals
3. **How big is it?** — Effect sizes (Cohen's d)

A complete analysis addresses all three. Let's examine each in turn.

2.3 The Null Hypothesis

To test whether our effect is real, we start with an assumption:

Null Hypothesis (H_0): The assumption that there is **no effect** or **no difference** in the population.

For the memory experiment:

- **H_0**: Word type has no effect on memory. Any observed difference is due to chance.
- In formal terms: μ_abstract = μ_concrete (the population means are equal)

2.4 The Alternative Hypothesis

The alternative (or research) hypothesis represents what we're actually interested in:

Alternative Hypothesis (H_1 or Ha): The prediction that there **is** an effect or difference.

For the memory experiment:

- **H_1** (two-tailed): Word type affects memory (μ_abstract ≠ μ_concrete)
- **H_1** (one-tailed): Concrete words are remembered better than abstract words (μ_concrete > μ_abstract)

2.5 The Logic of NHST

Null Hypothesis Significance Testing (NHST) follows this logic:

1. **Assume H_0 is true** (assume there's no real effect)
2. **Ask**: How likely is our observed data if H_0 were true?
3. **If the data is very unlikely under H_0**, reject H_0 and conclude there probably is an effect

> ! Key Insight
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> We never "prove" anything. We either reject H_0 (suggesting there's an effect) or fail to reject H_0 (the data doesn't provide enough evidence for an effect).

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3 Part 2: Understanding p-values

3.1 What Is a p-value?

p-value: The probability of obtaining results at least as extreme as the observed results, **assuming the null hypothesis is true**.

Let's unpack this:

- We calculate how extreme our data is
- We ask: "If there were truly no effect, how often would we see data this extreme just by chance?"
- The p-value answers this question

3.2 A Concrete Example

Imagine the memory experiment has a p-value of 0.03.

This means: **If** word type truly had no effect on memory, we would observe a difference as large as (or larger than) 1.4 words in only 3% of experiments, just by chance.

Because 3% is quite rare, we conclude that the null hypothesis is probably wrong — word type probably does affect memory.

3.3 The Significance Threshold (α)

We need a cutoff for deciding when p is "small enough" to reject H_0. This is called **alpha (α)**.

Alpha (α): The significance level — the threshold below which we reject the null hypothesis. Conventionally set at 0.05 (5%).

- If **$p < \alpha$** (typically $p < .05$): Reject H_0 — result is "statistically significant"
- If **$p \geq \alpha$** (typically $p \geq .05$): Fail to reject H_0 — result is "not statistically significant"

3.4 What p-values Do NOT Mean

⚠ Critical Misconceptions

The p-value is **NOT**:

1. The probability that H_0 is true
2. The probability that H_1 is false
3. The probability that the results occurred by chance
4. A measure of effect size or importance

A p-value only tells you how surprising your data would be IF H_0 were true. It says nothing about how likely H_0 actually is.

3.5 The Correct Interpretation

Wrong: “There’s a 3% chance this finding is due to chance.”

Correct: “If the null hypothesis were true, we’d see results this extreme only 3% of the time.”

This distinction matters! The p-value is calculated *under the assumption* that H_0 is true. It doesn’t tell you whether H_0 is actually true.

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4 Part 3: Type I and Type II Errors

4.1 Two Ways to Be Wrong

When we make decisions based on p-values, we can make two types of errors:

	H_0 is actually TRUE (no effect)	H_0 is actually FALSE (real effect)
Reject H_0	**Type I Error (False Positive)**	Correct Decision
Fail to reject H_0	Correct Decision	**Type II Error (False Negative)**

4.2 Type I Error (False Positive)

Type I Error: Rejecting H_0 when it’s actually true — claiming an effect exists when it doesn’t.

- The probability of a Type I error equals α (typically 5%)
- If you use $\alpha = .05$, you’ll falsely detect “effects” 5% of the time when there’s actually nothing there

DataLab Example: You conclude that concrete words are better remembered than abstract words, but actually the difference you observed was just due to random variation in who happened to be in each lab.

4.3 Type II Error (False Negative)

Type II Error: Failing to reject H_0 when it’s actually false — missing a real effect.

- The probability of a Type II error is denoted β (beta)
- **Power** $= 1 - \beta$ = the probability of correctly detecting a real effect

DataLab Example: Concrete words really are easier to remember, but your study had too few participants to detect this reliably. You conclude “no significant difference” and miss the real effect.

4.4 The Trade-off

You can’t minimise both errors simultaneously:

- Setting a stricter α (e.g., .01 instead of .05) reduces Type I errors but increases Type II errors
- Using larger samples reduces Type II errors without increasing Type I errors (this is why sample size matters!)

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5 Part 4: Confidence Intervals — How Certain Are We?

5.1 Beyond Point Estimates

A mean like 6.8 is a **point estimate** — our best single guess at the population value. But how confident are we in this estimate?

Confidence Interval (CI): A range of values that is likely to contain the true population parameter. A 95% CI represents the range within which we're reasonably confident the true value lies.

Instead of reporting: "Lab B mean = 6.8"

Report: "Lab B mean = 6.8, 95% CI [6.07, 7.53]"

This tells readers not just our estimate but how *precise* that estimate is.

5.2 Interpreting 95% Confidence Intervals

⚠ Common Misconception

A 95% CI does NOT mean "there's a 95% probability the true value is in this interval."

The correct interpretation:

> If we repeated this study many times and calculated a 95% CI each time, **95% of those intervals would contain the true population value**.

Once calculated, any specific CI either contains the true value or it doesn't — we just don't know which.

5.3 What CIs Tell Us

1. **The range of plausible values**: The true population mean is probably somewhere in this range
2. **Precision**: Narrower CI = more precise estimate (usually from larger samples)
3. **Uncertainty**: Wider CI = more uncertainty about the true value

5.4 CIs and Statistical Significance

Key rule: If the 95% CI for a difference **does not include zero**, the difference is statistically significant at $p < .05$.

For our memory study:

- Difference between labs = 1.4 words
- 95% CI for the difference = [0.42, 2.38]
- The CI does NOT include zero
- Therefore: $p < .05$ (significant difference)

If the CI had been [−0.20, 3.00], it would include zero, suggesting the difference might not be real.

5.5 Why Report CIs?

1. They provide more information than p-values alone
2. They show the range of plausible effects
3. Readers can judge whether the effect is practically meaningful
4. They're becoming required in many journals

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6 Part 5: Effect Sizes — How Big Is It?

6.1 Statistical Significance ≠ Practical Importance

⚠ Critical Distinction

A statistically significant result can be trivially small. With a large enough sample, even a 0.1-point difference could be "significant" — but is that meaningful?

This is why we need **effect sizes**.

6.2 Cohen's d

Cohen's d: A standardised measure of the difference between two means, expressed in standard deviation units.

$$d = \frac{\bar{X}_1 - \bar{X}_2}{s_{pooled}}$$

Effect sizes are independent of sample size. A $d = 0.50$ means the same thing whether you have 20 or 2,000 participants.

6.3 Interpreting Cohen's d

Cohen (1988) provided benchmarks:

d value	Interpretation	What it looks like
0.2	Small	Means overlap 85%
0.5	Medium	Means overlap 67%

d value	Interpretation	What it looks like
0.8	Large	Means overlap 53%

> **i Context Matters**
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> These benchmarks are rough guidelines. In some fields (e.g., clinical interventions), d = 0.3 is impressive. In others (e.g., perceptual differences), d = 0.3 might be trivial.

6.4 Calculating Cohen's d: Example

DataLab Memory Data:

- Lab A: M = 5.4, SD = 1.8
- Lab B: M = 6.8, SD = 2.0

First, calculate pooled SD:

$$s_{pooled} = \sqrt{\frac{(SD_1)^2 + (SD_2)^2}{2}} = \sqrt{\frac{1.8^2 + 2.0^2}{2}} = \sqrt{\frac{3.24 + 4.0}{2}} = \sqrt{3.62} = 1.90$$

Then calculate d:

$$d = \frac{6.8 - 5.4}{1.90} = \frac{1.4}{1.90} = 0.74$$

This is a **medium-to-large effect** — the lab manipulation had a meaningful impact on memory.

6.5 Why Effect Sizes Matter

1. **Practical importance**: Tells you if the effect is big enough to care about
2. **Comparability**: Allows comparison across different studies and measures
3. **Meta-analysis**: Effect sizes are combined across studies to estimate true effects
4. **Complete picture**: p-values tell you IF there's an effect; d tells you HOW BIG

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7 Part 6: The Independent Samples t-test

7.1 When to Use It

The **independent samples t-test** compares means from two **different** groups of participants.

Use it when:

- You have two separate groups
- Each participant is in only one group (between-subjects design)
- You want to compare the groups on a continuous outcome

DataLab Example: Comparing memory scores between Lab A (abstract words) and Lab B (concrete words) — different students in each lab.

7.2 The t-statistic

The t-test produces a **t-statistic** that captures the ratio of signal to noise:

$$t = \frac{\text{Difference between means}}{\text{Standard error of the difference}}$$

In words: **signal ÷ noise**

- **Larger t** = bigger difference relative to variability = more evidence against H_0
- **Smaller t** = difference is small relative to variability = weaker evidence

7.3 Worked Example: Independent t-test

DataLab Data:

- Lab A (abstract): *M* = 5.4, *SD* = 1.8, *n* = 30
- Lab B (concrete): *M* = 6.8, *SD* = 2.0, *n* = 30

Step 1: Calculate the pooled standard error

$$SE_{diff} = \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}} = \sqrt{\frac{1.8^2}{30} + \frac{2.0^2}{30}} = \sqrt{0.108 + 0.133} = 0.49$$

Step 2: Calculate t

$$t = \frac{\bar{x}_2 - \bar{x}_1}{SE_{diff}} = \frac{6.8 - 5.4}{0.49} = \frac{1.4}{0.49} = 2.86$$

Step 3: Degrees of freedom

df = $n_1 + n_2$ - 2 = 30 + 30 - 2 = 58

Step 4: Look up p-value

With df = 58, t = 2.86 gives p = .006

Step 5: Calculate effect size

d = (6.8 - 5.4) / 1.90 = 0.74

Step 6: Make a decision

Since p = .006 < .05, we reject H_0. The difference is statistically significant, with a medium-to-large effect.

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8 Part 7: The Paired Samples t-test

8.1 When to Use It

The **paired samples t-test** compares two measurements from the **same** participants.

Use it when:

- The same people provide both scores (within-subjects design)
- You want to compare before vs after, or condition 1 vs condition 2 within the same person

DataLab Examples:

- Pre-session mood vs post-session mood
- Dominant hand accuracy vs non-dominant hand accuracy

8.2 Why "Paired"?

Each person has a pair of scores. We're interested in **within-person change** — did individuals improve, decline, or stay the same?

8.3 The Key Difference from Independent t-test

Instead of comparing two separate groups, we:

1. Calculate the **difference score** for each person
2. Test whether the **mean difference** is significantly different from zero

Paired t-test logic: If there's no effect, the average difference should be zero. We test whether our observed mean difference is far enough from zero to be statistically significant.

8.4 Worked Example: Paired t-test

DataLab Data (Mood pre-post):

For 30 participants:

- Mean pre-mood: 5.3
- Mean post-mood: 6.5
- Mean difference: *M*_diff = 1.2
- SD of differences: *SD*_diff = 1.8

Step 1: Calculate standard error of the mean difference

$$SE_{diff} = \frac{SD_{diff}}{\sqrt{n}} = \frac{1.8}{\sqrt{30}} = \frac{1.8}{5.48} = 0.33$$

Step 2: Calculate t

$$t = \frac{\bar{d}}{SE_{diff}} = \frac{1.2}{0.33} = 3.64$$

Step 3: Degrees of freedom

df = n - 1 = 30 - 1 = 29

Step 4: Look up p-value

With df = 29, t = 3.64 gives p = .001

Step 5: Calculate effect size

$$d = \frac{\bar{d}}{SD_{diff}} = \frac{1.2}{1.8} = 0.67$$

Step 6: Make a decision

Since p = .001 < .05, we reject H_0. Mood significantly improved after the session.

8.5 Why Paired Tests Are More Powerful

Paired tests control for individual differences. Some people are naturally happier; some have better memories. By comparing each person to themselves, we remove this "noise," making it easier to detect true effects.

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9 Part 8: Complete APA Reporting

9.1 The Formula

APA format for t-tests:

t(df) = X.XX, *p* = .XXX, *d* = X.XX, 95% CI [X.XX, X.XX]

All five components:

1. **t** = test statistic
2. **df** = degrees of freedom (in parentheses)
3. **p** = exact p-value (or p < .001 if very small)
4. **d** = Cohen's d effect size
5. **95% CI** = confidence interval for the difference

9.2 Complete Example: Independent t-test

> An independent samples t-test revealed that participants who studied concrete words (*M* = 6.8, *SD* = 2.0) recalled significantly more words than those who studied abstract words (*M* = 5.4, *SD* = 1.8), *t*(58) = 2.86, *p* = .006, *d* = 0.74, 95% CI [0.42, 2.38].

Notice this includes:

- Descriptive statistics for both groups
- The t-statistic with degrees of freedom
- The exact p-value
- The effect size (Cohen's d)

- The 95% confidence interval for the difference

9.3 Complete Example: Paired t-test

> A paired samples t-test showed that participants' mood ratings were significantly higher after the DataLab session (M = 6.5, SD = 1.4) compared to before (M = 5.3, SD = 1.6), $t(29) = 3.64$, $p = .001$, $d = 0.67$, 95% CI [0.54, 1.86].

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10 Part 9: Choosing the Right t-test

10.1 Decision Guide

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Are the two groups of scores from the same people?
                           |
            ┌──────────────┴──────────────┐
           Yes                           No
            ↓                             ↓
     PAIRED t-test           INDEPENDENT t-test
```

10.2 DataLab Applications

Comparison	Design	Test
Lab A vs Lab B memory	Between-subjects	Independent t-test
Pre vs Post mood	Within-subjects	Paired t-test
Dominant vs Non-dominant hand	Within-subjects	Paired t-test
Males vs Females on any task	Between-subjects	Independent t-test

This connects directly to Week 13: your **design choice** determines your **test choice**.

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11 Summary

Concept	Key Point
Null hypothesis (H_0)	Assumes no effect — what we test against
Alternative hypothesis (H_1)	Predicts an effect exists
p-value	Probability of data this extreme if H_0 were true
α (alpha)	Significance threshold (typically .05)
Type I error	False positive — claiming an effect that doesn't exist

Concept	Key Point
Type II error	False negative — missing a real effect
Confidence interval	Range of plausible values for the true parameter
Effect size (d)	Standardised measure of how big the effect is
Independent t-test	Compares means from different groups
Paired t-test	Compares means from the same people

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12 The Complete Analysis Checklist

When reporting results, ensure you include:

- ☐ Descriptive statistics (M, SD) for each group/condition
- ☐ The appropriate test (independent or paired t-test)
- ☐ Test statistic (t) with degrees of freedom
- ☐ Exact p-value
- ☐ Effect size (Cohen's d)
- ☐ 95% confidence interval
- ☐ Interpretation in words

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13 Check Your Understanding

Before the lecture, make sure you can answer these questions:

1. What is the null hypothesis and why do we test against it?
2. In your own words, what does a p-value of 0.02 mean?
3. What's the difference between a Type I and Type II error?
4. How do you interpret a 95% confidence interval?
5. What does Cohen's d tell us that p-values don't?
6. When would you use an independent t-test vs a paired t-test?
7. What does it mean if a result is "not statistically significant"?
8. What are the five components of complete APA reporting for t-tests?

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14 Further Reading

For additional depth:

- Cumming, G. (2012). *Understanding the New Statistics*. Chapters on NHST, CIs, and effect sizes.
- Coolican, H. (2017). Chapters 17-18: Significance testing and t-tests. VLeBooks
- Lakens, D. (2013). Calculating and reporting effect sizes. *Frontiers in Psychology, 4*, 863.